KAIZEN MATHS

Lesson starter

Arithmetic and Geometric Sequences and Series

Geometric Sequences and Series

Name _____

S1

For a geometric sequence with $a = 4$ and $r = \frac{2}{3}$, find u_8.

S2

In a geometric sequence, $u_4 = -32$ and $u_7 = 256$. Find r and a.

S3

For a geometric sequence with $a = 4$ and $r = -2$, find S_5.

S4

For a geometric series with $a = 6$ and $r = \frac{3}{4}$, find S_∞.

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Exit ticket

Arithmetic and Geometric Sequences and Series

Geometric Sequences and Series

Name _____

E1

Evaluate $\sum_{k=1}^{4} 1(\frac{2}{3})^k$ as a fraction.

E2

For a geometric sequence with $a = 8$ and $r = 3$, find u_9.

E3

In a geometric sequence, $u_4 = 24$ and $u_6 = 96$. Find r and a.

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Teacher answers & working

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S1

For a geometric sequence with $a = 4$ and $r = \frac{2}{3}$, find u_8.

Answer:

$u_8 = \frac{512}{2187}$

Working:

$u_n = ar^{n-1}$

$u_8 = 4(\frac{2}{3})^7$

$\boxed{u_8 = \frac{512}{2187}}$

S2

In a geometric sequence, $u_4 = -32$ and $u_7 = 256$. Find r and a.

Answer:

$r = -2, \quad a = 4$

Working:

$u_n = ar^{n-1}$

$\frac{u_7}{u_4} = r^3$

$r = -2$

$a = \frac{u_4}{r^3} = 4$

S3

For a geometric sequence with $a = 4$ and $r = -2$, find S_5.

Answer:

$S_5 = 44$

Working:

$S_n = \frac{a(1 - r^n)}{1 - r}$

$S_5 = \frac{4(1 - (-2)^5)}{1 - -2}$

$\boxed{S_5 = 44}$

Teacher answers & working

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S4

For a geometric series with $a = 6$ and $r = \frac{3}{4}$, find S_∞.

Answer:

$S_\infty = 24$

Working:

$|r| < 1 \Rightarrow S_\infty = \frac{a}{1-r}$

$S_\infty = \frac{6}{1-\frac{3}{4}}$

$\boxed{S_\infty = 24}$

E1

Evaluate $\sum_{k=1}^{4} 1(\frac{2}{3})^k$ as a fraction.

Answer:

$= \frac{130}{81}$

Working:

$\sum_{k=p}^{q} Ar^k = Ar^p \frac{1-r^{q-p+1}}{1-r}$

$= 1(\frac{2}{3})^1 \frac{1-(\frac{2}{3})^4}{1-\frac{2}{3}}$

$\boxed{= \frac{130}{81}}$

E2

For a geometric sequence with $a = 8$ and $r = 3$, find u_9.

Answer:

$u_9 = 52488$

Working:

$u_n = ar^{n-1}$

$u_9 = 8(3)^8$

$\boxed{u_9 = 52488}$

Teacher answers & working

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E3

In a geometric sequence, $u_4 = 24$ and $u_6 = 96$. Find r and a.

Answer:

$r = 2, \quad a = 3$

Working:

$u_n = ar^{n-1}$

$\frac{u_6}{u_4} = r^2$

$r = 2$

$a = \frac{u_4}{r^3} = 3$